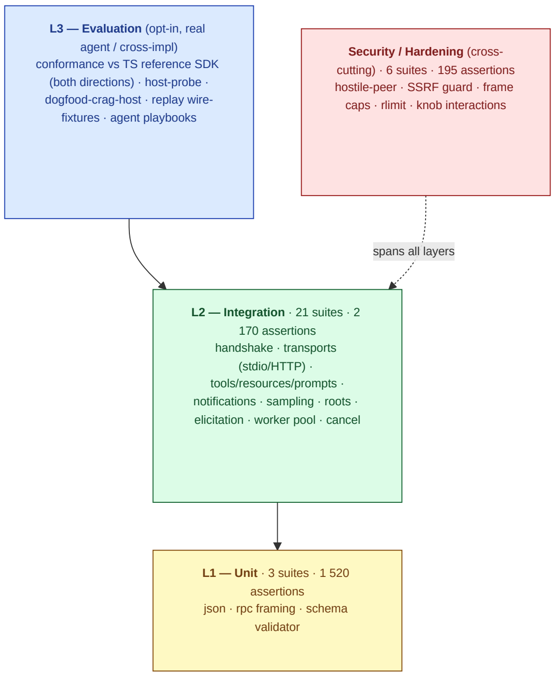
cMCP — Testing Overview

A visual map of cMCP's test and quality posture, framed against the three-layer MCP test pyramid (guide Ch.3.1), the load/resilience metrics (Ch.4), and the MCP security controls (Ch.6).

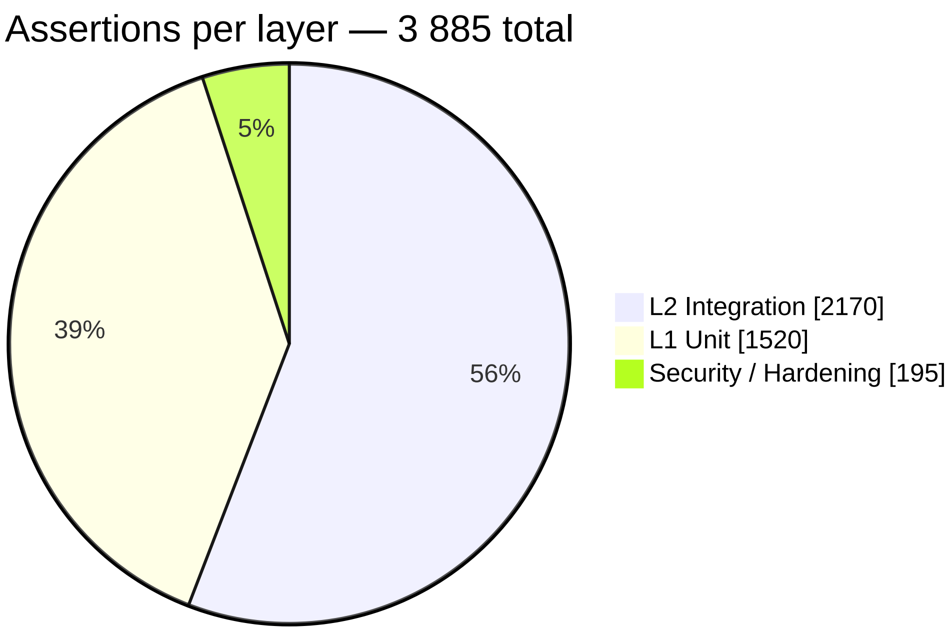
All numbers below are from the hermetic `make test` run on 2026-06-08: **3 885 assertions across 30 test binaries**, green under valgrind + ASan/UBSan + TSan.

1. The test pyramid (guide Ch.3.1)

cMCP's suites mapped onto the guide's Unit → Integration → Evaluation hierarchy, with a security/hardening axis that cuts across all layers.



Assertion distribution (`make test`)



Layer	Suites	Assertions	Relative
L1 — Unit	3	1 520	
L2 — Integration	21	2 170	
Security / Hardening	6	195	
L3 — Evaluation	opt-in	cross-impl*	—

* L3 is opt-in (`make conformance` / `make replay` , needs Node+network): 32 assertions cMCP-client-vs-TS-server, 8 checks TS-client-vs-cMCP-server, plus the stateful `host-probe` and `dogfood-crag-host` consumer probes.

L1 — Unit

Suite	Assertions
test_rpc	800
test_schema	600
test_json	120

L2 — Integration

Suite	Assertions
test_stdio_roundtrip	807
test_client_helpers	224
test_resources_prompts	133
test_http_server	110
test_tools	108
test_client_server	106
test_lifecycle	99
test_elicitation	80
test_fs_server	78
test_logging	53
test_http_client	51
test_roots	50
test_notifications	47
test_structured_output	41
test_workers	38
test_crag_cutoff	38
test_sampling	37
test_host_cancel_progress	30
test_ping	14
test_pagination	13
test_http_auth	13

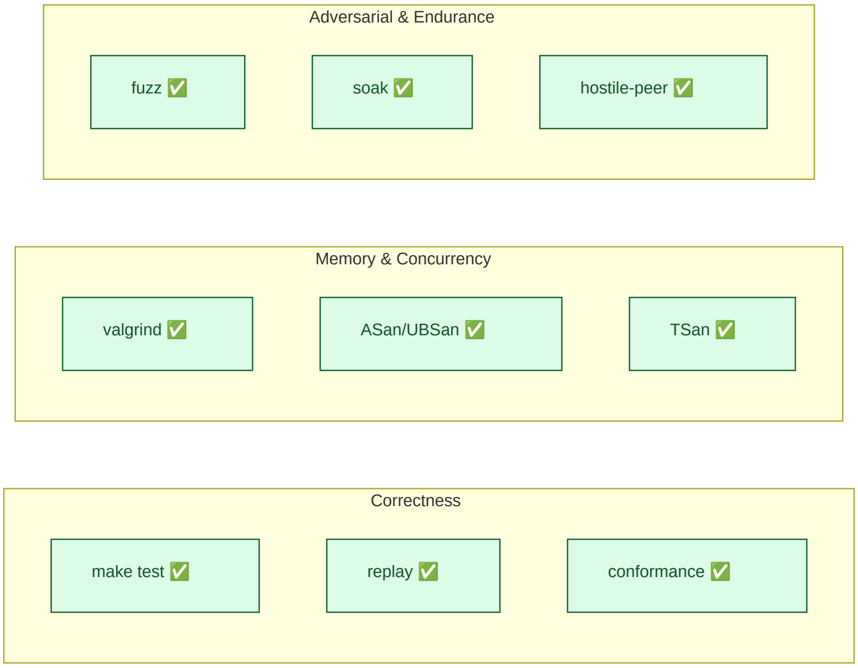
Security / Hardening

Suite	Assertions
test_http_ssrf_guard	76
test_hostile_peer	71
test_hardening_cross_knob	22
test_stdio_frame_cap	12
test_tee_frame_cap	10
test_handler_rlimit	4

2. Quality gates

Every gate below is wired into CI. ✅ = green as of 2026-06-08.

Gate	Command	What it proves	Status
Unit + integration	`make test`	3 885 assertions / 30 binaries	✅
Memory safety	`make valgrind`	leak-free, no invalid access	✅
ASan + UBSan	`make test-asan`	no heap/UB errors	✅
ThreadSanitizer	`make test-tsan`	no data races	✅
Fuzzing	`make fuzz-smoke`	4 libFuzzer harnesses (json/rpc/schema/http)	✅
Wire regression	`make replay`	captured-transcript replay gate	✅
Soak (stdio+HTTP)	`make soak` / `soak-http`	no drift over long runs	✅
Cross-impl conformance	`make conformance`	matches TS reference SDK, both roles	✅
Spec-drift watch	`make check-spec-drift`	pinned protocol vs upstream	✅



3. cMCP vs the state of the art (guide Ch.4)

Measured head-to-head against the **official reference SDKs**: the same cMCP client, the same `tools/call echo` workload over stdio, with the subprocess server swapped per row (`make bench-compare` , this machine, 2026-06-08, 10 000 iterations after 1 000 warmup). Per the guide's metric priority (Ch.4.2), latency sits far inside the sub-millisecond band that disappears inside an LLM inference cycle.

Throughput — calls/s (higher is better)

Server	calls/s	vs cMCP
cMCP (C11)	**43 494**	**1.0×**
TypeScript SDK (Node)	8 035	0.18×
Python SDK (FastMCP)	974	0.02×

→ cMCP sustains **5.4× the TypeScript SDK** and **45× the Python SDK**.

Per-call latency — µs (lower is better)

Server	p50	p99	mean
cMCP	**21**	**31**	**22**
TypeScript SDK	95	189	123
Python SDK	994	1 315	1 025

Idle memory footprint — RSS (lower is better)

Server	idle RSS	vs cMCP
cMCP (328 KB static binary)	**2.2 MB**	**1.0×**
Python SDK (FastMCP)	60 MB	27×
TypeScript SDK (Node)	74 MB	33×

Why the gap

	cMCP	TypeScript SDK	Python SDK
Language / runtime	C11, native	Node.js	CPython
Third-party deps	none*	npm (`sdk`, `zod`)	pip (`mcp`/FastMCP)
Server artifact	328 KB static binary	runtime + `node_modules`	interpreter + venv
Conformance	cross-checked vs the TS reference SDK, both wire roles	reference	—

* libcurl is linked only by the HTTP *client*; a stdio server carries zero third-party dependencies.

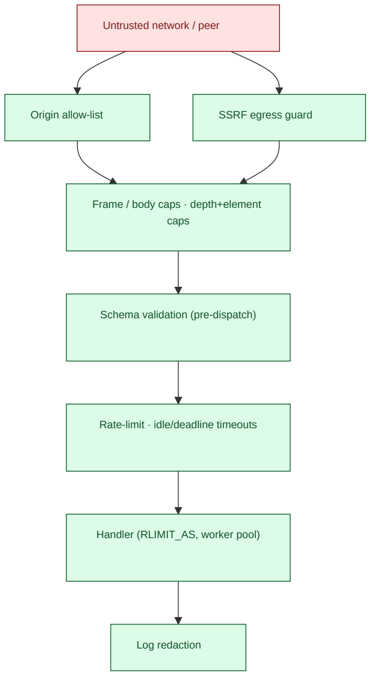
- **cMCP's pure in-process ceiling** (no subprocess, inline pipe pair) is **50 487 calls/s at p50 = 19 µs** — the cross-SDK rows above each pay one subprocess hop so all three are measured identically.
- **Resilience knobs (Ch.4.1 spike/idle traffic):** HTTP accept-rate token bucket, per-recv idle timeout + cumulative deadline (slowloris), stdio/HTTP frame caps (OOM).

  Numbers vary per machine/kernel/build; regenerate with `make bench-compare`. Wire-compatibility with the reference SDK is separately gated by `make conformance` (both directions).

4. Security controls (guide Ch.6)

How cMCP's hardening maps onto the guide's MCP threat catalogue. TLS is deliberately terminator-only (reverse proxy); the deployment threat model is documented in the architecture notes.

Guide threat (Ch.6)	cMCP control	Env knob	Test
SSRF via discovery (6.1)	egress guard on resolved peer (rejects metadata/link-local/RFC1918/CGNAT/ULA); DNS-rebinding-safe	`CMCP_HTTP_ALLOW_PRIVATE`	test_http_ssrf_guard
DNS rebinding (6.1)	`Origin` allow-list on HTTP server	`CMCP_HTTP_ALLOWED_ORIGINS`	test_http_server
Insecure deserialization (6.1/6.3)	schema validation **before** dispatch; depth + element caps	`CMCP_JSON_MAX_DEPTH` / `_ELEMENTS`	test_schema, test_hostile_peer
Memory-bomb / OOM (4)	stdio frame cap, HTTP body cap, handler `RLIMIT_AS`	`CMCP_STDIO_MAX_FRAME`, …	test_stdio_frame_cap, test_handler_rlimit
Slowloris / spike (4)	accept-rate token bucket, idle + deadline timeouts	`CMCP_HTTP_ACCEPT_RATE`, `_IDLE_TIMEOUT_MS`, `_DEADLINE_MS`	test_hardening_cross_knob
Credential leakage (6)	log redactor over credential-shaped keys	`CMCP_LOG_REDACT`	test_http_auth, test_logging
Malformed / hostile peer	bounded parsers, fail-closed framing	—	test_hostile_peer



Regenerate the assertion counts with `make test`; this document is a point-in-time snapshot (2026-06-08).